

JEE MAIN REVISION MASTER: RAY OPTICS 2026 5 QUESTIONS 8.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 38 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Ray Optics - Core Application	38 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Ray Optics - Core Application (Very High Yield — 38 PYQs)

Core Principle: Key problem pattern in Kcet Ray Optics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "With reference to the figure shown below, match the following: List-I List-II (a) Angle of reflection (i) 60° (b) Value of α ..." [Ans: A]
- [2026 Shift] Q#2 (MCQ): "The critical angle for a monochromatic light going from medium A to medium B is θ . If the speed of light in medium A is V , then t ..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is

- (A) $\frac{\sqrt{3}}{2}$
- (B) $(3/2)$
- (C) $\sqrt{3}$
- (D) $\sqrt{2}$

Correct Answer: None

Solution Analysis: For an equilateral prism, the angle of prism is $A=60^\circ$. From the graph of angle of deviation δ versus angle of incidence i , the lowest value of deviation is the minimum deviation. Here, the graph shows $\delta_m=60^\circ$. The refractive index μ of the prism material is given by $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$. Now put $A=60^\circ$ and $\delta_m=60^\circ$ in the formula. $\mu = \frac{\sin\left(\frac{60^\circ+60^\circ}{2}\right)}{\sin\left(\frac{60^\circ}{2}\right)}$ $\mu = \frac{\sin(60^\circ)}{\sin(30^\circ)}$ Using $\sin 60^\circ = \frac{\sqrt{3}}{2}$ and